
FILIPPO ANDREA MARIANI

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PROFESSIONAL SUMMARY

I am a Mechanical Engineer with a specialization in Mechatronics and Robotics, graduated from **Politecnico di Milano** with a final grade of 110/110 (GPA 4.0/4.0). I am highly motivated, enjoy collaborative team environments, and have a strong passion for robotics and automation. My primary interests lie in the design and control of robotic systems, particularly biomedical devices and exoskeletons that assist individuals with impaired mobility.

I completed my Master's Thesis at **Harvard University**, working in Prof. Patrick Slade's lab on a supernumerary robotic exoskeleton. My work focused on developing a Machine Learning-based control system to assist users with multiple mobility tasks.

Currently, I am working at **R4P S.r.l.**, a research and development company, as a Robotics Engineer. My role involves enhancing control systems and automating industrial robotics applications, including both automated production lines and autonomous robotic systems.

EXPERIENCE

Robotics Engineer, 01/2025 - Current

R4P S.r.l. – Milan, Italy

- Engineered software solutions for robotics systems to automate tasks and enhance operational efficiency.
- Collaborated with cross-functional teams to integrate robotics solutions into existing workflows.
- Assisted in programming PLC systems for Siemens and Omron. Supported troubleshooting and maintenance of automated processes.
- Executed HMI programming utilizing Optix software to enhance user interface functionality.

Academic Tutor (Computer Science B Class), 09/2024 - 02/2025

Politecnico Di Milano – Milan, Italy

- Collaborated with faculty to align tutoring approaches with curriculum objectives and student learning outcomes.
- Monitored student progress through assessments and provided constructive feedback for improvement.
- Coordinated and delivered engaging laboratory sessions focused on practical applications of MATLAB and C programming languages.
- Supervised student engagement during class sessions and examinations to maintain academic integrity.

Mechanical Engineering (track: Mechatronics and Robotics), 12/2025

Politecnico Di Milano - Milan, Italy

Thesis Project

Final grade: 110/110 - GPA equivalent 4.0/4.0

Simulation-Informed Human-in-the-Loop Optimization for Supernumerary Robotic Assistance in Activities of Daily Living

Harvard University - School of Engineering and Applied Sciences (SEAS) | March 2025 - September 2025

In collaboration with Prof. Patrick Slade and the Ability Lab

GPA Equivalent: 4.0/4.0

- Conducted a six-and-a-half-month research period at Harvard University (SEAS) within Prof. Patrick Slade's laboratory.
- Contributed to the development of a supernumerary robotic exoskeleton aimed at assisting elderly individuals and people with motor impairments.
- Developed a Machine Learning algorithm for real-time movement phase estimation and task segmentation across multiple activities.
- Adapted the exoskeleton's control logic and Human-in-the-Loop Optimization (HILO) framework to support three activities (sit-to-stand, object lifting, and stair step-up) while maintaining a unified control architecture to assess system and pipeline generalizability.
- Conducted a healthy human subject testing campaign to experimentally validate system performance.
- Publication currently in preparation in collaboration with Prof. Slade and Dr. Cho.

Relevant Projects

Control of a Linear Inverted Pendulum

Mechanical Design and Control Project | Politecnico di Milano | Sept 2024 - Feb 2025

GPA Equivalent: 4.0/4.0

- Developed and implemented a swing-up control algorithm for a cart-mounted pendulum.
- Designed stabilization control for both equilibrium positions.
- Implemented the system using Arduino integrated with a Simulink environment.

3D Reconstruction of a Building with Drones

Mechanical Design Project | Politecnico di Milano | Sept 2024 -Feb 2025

GPA Equivalent: 4.0/4.0

- Acquired aerial imagery using drones and performed photogrammetric processing.
- Reconstructed a 3D model of the area using Agisoft Metashape software.

Creation of a VR Environment for Sports Applications

Mechanical Design Project | Politecnico di Milano | Sept 2024 - Feb 2025

GPA Equivalent: 4.0/4.0

- Developed an augmented/virtual reality environment to enhance sports training for individuals recovering from injuries.
- Implemented the system using Unity and Meta Quest 3 VR headsets.

Industrial Robotics

Mechanical Design Project | Università di Pavia | Sept 2024 - Feb 2025

GPA Equivalent: 4.0/4.0

- Designed a 4-DOF robotic manipulator using CAD software.
- Developed kinematic and dynamic models in MATLAB and Simscape.
- Performed trajectory planning and performance analysis for a palletizing task.
- Programmed and implemented control on a real TM collaborative robot at the Pavia Laboratory.

Data Analysis Project

Mechanical Engineering Project | Politecnico di Milano | Feb 2024 - Jun 2024

GPA Equivalent: 4.0/4.0

- Conducted output-only modal analysis on structural data from Palazzo Regione Lombardia.
- Applied the Ibrahim Time Domain method for system identification.

Bachelor of Science: Mechanical Engineering, 09/2023

Politecnico Di Milano - Milan, Italy

- **Final Grade:** 106/110 (Italian scale; approx. 3.85/4.0 U.S. GPA equivalent)
- **Thesis Statement:** "Design and validation of a front-loading washing machine".

Relevant Projects

Bicycle Frame Structural Analysis

Grade: [30/30] - **GPA Equivalent:** [4.0/4.0]

- Designed and structurally verified a bicycle frame under multiple loading conditions.
- Performed dimensioning calculations based on static and dynamic load cases.
- Analyzed critical scenarios including acceleration and deceleration phases.
- Applied principles of solid mechanics and structural analysis to optimize frame performance.

High School diploma: 07/2020

Liceo Scientifico Don Carlo Gnocchi - Carate Brianza

Final grade: 100/100

SKILLS

Programming languages: Proficient knowledge:
C, MATLAB, Python, Simulink

Intermediate knowledge: ROS, C#

Software: CAD: CATIA V5, Inventor, Solidworks,
Solidedge

PLC: Siemens TIA Portal, Sysmac Studio

Unity, NI Vision Assistant

Spreadsheets: MS Office package

3D Printing Technologies: Bambulab 3D printers

LANGUAGE

Italian: mother tongue

English: fluent both written and spoken FIRST
Certificate, C1

 Advanced (C1)

PERSONAL AND WORKING EXPERIENCES

I was responsible for organizing various charity fundraisers for the AVSI non-profit organization. I hold certificates in basic, intermediate, and advanced levels of wine tasting. I enjoy fishing, skiing and filming with drones.